

# NET731-FA3

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# PROJECT OVERVIEW

I CREATED A SECURE AND SCALABLE CLOUD  
INFRASTRUCTURE FOR TECHBRIDGE SOLUTIONS.

DESIGNED TO PROVIDE HIGH AVAILABILITY & ZERO TRUST  
SECURITY WITH AUTOMATED COST CONTROLS.

THE OUTCOME WAS A INTEGRATED AZURE ECOSYSTEM  
WHICH USES VENT'S AND RBAC

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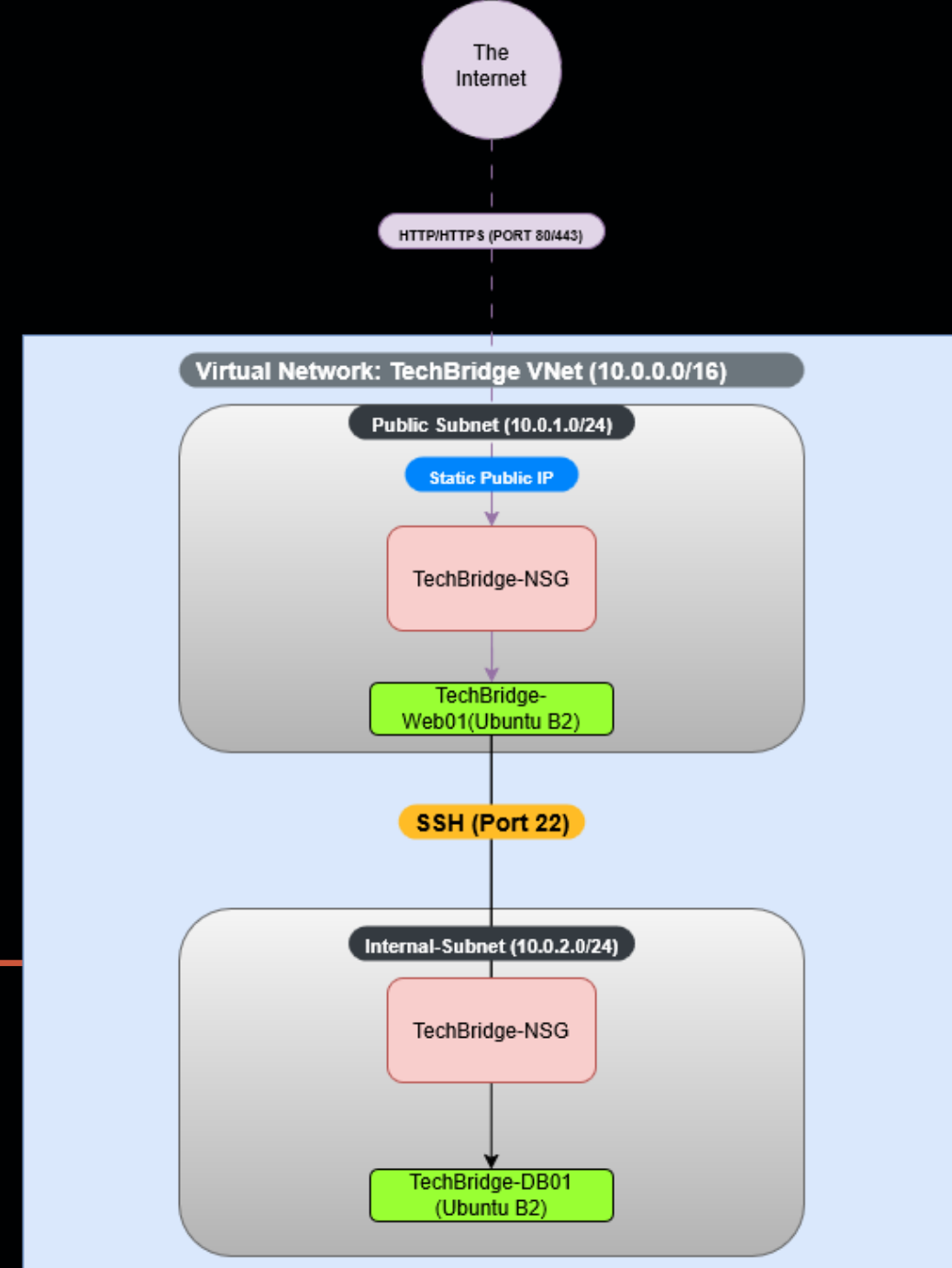
# ARCHITECTURE DIAGRAM

THE ENVIRONMENT USES A 10.0.0.0/16 VNET WITH TWO SUBNETS

\*PUBLIC-SUBNET: WHICH ACTS AS DMZ WITH WEB01

\*INTERNAL-SUBNET: PRIVATE SUBNET FOR THE BACKEND VM (DB01)

AND LASTLY A UNIFIED NSG THAT COVERS BOTH SUBNETS



Microsoft Defender for Cloud

Azure Monitor

Storage Account: techbridgestorage (LR8)

# TECHNICAL DECISIONS

I USED B2'S SKU FOR THEIR COST EFFICIENCY (B1'S WAS NOT AVAILABLE) AND PROVIDE 750 HOURS/MONTH ON THE AZURE FREE TRIAL WHICH SAVES THE \$200 CREDITS.

I USED LOCALLY REDUNDANT STORAGE (LRS) TO MINIMIZE SPENDING AS HIGH AVAILABILITY WASN'T A REQUIREMENT.

A AUTO-SHUTDOWN WAS CONFIGURED DAILY FOR 7PM TO PREVENT CREDIT DRAIN.



# COST MANAGEMENT

The screenshot displays the Microsoft Azure portal's Cost Management + Billing section. The left-hand navigation pane lists various services and resources, with 'Budgets' highlighted under the 'Reporting + analytics' category. The main content area is titled 'Cost Management: Aiden Horn | Budgets' and shows a table of existing budgets. A notification banner at the top right confirms that a budget named 'Capstone-Budget' has been successfully created. The table lists the following details for the 'Capstone-Budget':

| Name            | Scope                 | Reset period | Creation date | Expiration date | Budget     | Forecasted |
|-----------------|-----------------------|--------------|---------------|-----------------|------------|------------|
| Capstone-Budget | 1f643017-c53f-5267... | Monthly      | 01/05/2026    | 30/04/2028      | US\$100.00 |            |

# CHALLENGES AND SOLUTIONS

CHALLENGE: AVAILABILITY ERROR ON B1'S DUE TO LACK OF RESOURCES IN AZURE SOUTH-AFRICA-WEST REGION

SOLUTION: I USED THE B2'S INSTEAD AS IT FREE TIER SERVICES ELIGIBLE JUST LIKE B1'S AND WAS BASICALLY THE SAME WITH THE SAME AMOUNT OF VCPU'S AND RAM.

CHALLENGE: NOT BEING ABLE TO USE CTU STUDENT ACCOUNT AS ACCESS TO ENTRA ID WAS DENIED

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SOLUTION: CREATED A NEW PRIVATE AZURE ACCOUNT WITH \$200 FREE CREDIT

# LESSONS LEARNED

A SKILL LEARNED WAS USING IAAS (INFRASTRUCTURE AS A SERVICE).

I ALSO LEARNT ABOUT RBAC/IAM GOVERNANCE THROUGH LEAST PRIVILEGE .

LASTLY I LEARNT HOW TO TROUBLESHOOT RESOURCES IN THE CLOUD.

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